Probability and Statistics

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November 17, 2025

Content: Probability and Statistics & Reliability

Content

1. Recall on set Theory
2. **Counting Techniques:** Permutation and combination.
3. **Probability:** Basic concepts and definition of probability, Conditional probability.
4. **Probability distributions:** Discrete distributions(e.g. binomial and Poisson distributions) and Continuous distribution (e.g. Normal Distribution).
5. Sampling Method
6. Reliability Analysis

Counting Techniques

Permutations and Combinations

Introduction:

- Combinations and permutations are used in both statistics and probability, and they in turn, involve operations with factorial notation.
- Therefore combinations, permutations, and factorial notation are discussed in this section.
- This section aims at teaching a method of approach to certain problems involving arrangements and selections.
- This section helps us to count all elements of finite set; for counting all elements of finite set **the injection, surjection and Cartesian product** play important role.

Cartesian product

Consider A and B as the finite sets. A Cartesian product between A and B is defined as:

$$A \times B = \{(a, b) : a \in A \, and \, b \in B\}$$

In general,

$$\begin{aligned} A_1 \times A_2 \times \ldots \times A_n &= \{(a_1, a_, \ldots, a_n) : a_i \in A_i \, for \, i = 1, 2, \ldots, n \\ \#(A \times B) &= \#A.\#B \\ \#(A_1 \times A_2 \times \ldots \times A_n) &= \prod \#A_i \end{aligned}$$

Example

How many words of three alphabet can you form using the alphabet letters such that the first and third are consonants and the second is vowel.

Solution

Consider $\# C = 20$ and $\# V = 6$.

$$\#(C \times V \times C) = 20 \times 6 \times 20 = 2400 \; words$$

The union of 2 sets

Consider A and B as the finite sets.

$$
\begin{aligned}
A \cup B &= \{x \, such \, that \, x \in A \, or \, x \in B\} \\
\#(A \cup B) &= \#A + \#B - \#(A \cap B) \\
\#(A \cup B) &= \#A + \#B \, if \, A \cap B = \emptyset \\
\#\emptyset &= 0 \\
\#U &= n
\end{aligned}
$$

Arrangement with repetition

Definition: If we take p elements in n elements of E one after another with repetition means that before taking the second object in E we must put the first taken into E; we continue this process until we take p elements is called **arrangement with repetition.**

Notation:

$$\#\left(E^p\right) = \left(\# E\right)^p = n^p$$

Example

How many numbers can you write composed by 3 digits numbers different to zero?

Solution

$$E = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}, \ \#E = 9$$

We obtain $9^3 = 729$ numbers.

Arrangement without repetition

Definition: If we take r elements in n elements of E one after another without repetition means that taking one after the other in E; we continue this process until we take p elements is called **arrangement without repetition.**

Notation:

$$nPr = \frac{n!}{(n-r)!}$$

Example 1

How many permutations of six objects taken two a time can be made?

Answer:

$$6P2 = \frac{6!}{(6-2)!} = \frac{6 \times 5 \times 4!}{4!} = 30$$

Example 2

How many four-digit can be formed from the digits 2, 3, 4, 5, 6, and 7 without repetition?

Answer:

$$6P4 = \frac{6!}{(6-4)!} = \frac{6 \times 5 \times 4 \times 3 \times 2!}{2!} = 360$$

Particular cases

Permutation (bijection)

We call permutation of finite set E of $\#E = n$, all arrangement at n elements of E.

$$nPn = n!$$

Example

How many possibilities can you arrange 4 cars into a parking of 4 places?

Answer:

$$4P4 = 4! = 4 \times 3 \times 2 \times 1 = 24 \, possibilities$$

Permutation with repetition

The number of arrangements of n items, where there are k groups of like items of size $r_1, r_2, r_3, \ldots, r_k$, respectively, is given by

$$\frac{n!}{r_1!.r_2!.r_3! \ldots r_k!}$$

Example

How many different arrangements of the letters in the word ROOM can be made?

Answer:

$$n = 4,\ r_1 = 1,\ r_2 = 2,\ r_3 = 1$$

$$\frac{4!}{1!2!1!} = \frac{4!}{2!} = 12\, Arrangement$$

Combinations

- A combination is defined as a possible selection of a certain number of objects taken from a group without regard to the order.
- Consider E non-empty finite set of cardinal n and r is a natural number such that

$$0 \leq r \leq n.$$

We call combination at r elements of E all part of r elements of E.

$$\binom{n}{r} = nCr = \frac{n!}{r!(n-r)!}$$

Example

How many possible 14 students can form a committee of 6 students?
Each committee is a party of 6 students of the set of 14 students
Answer:

$$\binom{14}{6} \quad = \quad nCr = \frac{14!}{6!(14-6)!} = 3003$$

Probability notion

Introduction and terminology

- **Probability, (probability theory),** branch of mathematics that deals with measuring or determining quantitatively the likelihood that an event or experiment will have a particular outcome.
- It studies random experiment.
- Probability is based on the study of permutations and combinations and is the necessary foundation for statistics.

Probability Theory

Meaning of Probability

Assume that an experiment can be repeated many times, with each repetition called a **trial**, and assume that one or more outcomes can result from each trial, then the probability of a given outcome is the number of times that outcome occurs divided by the total number of trials. If the outcome is sure to occur, it has a probability of 1; if an outcome can not occur, its probability is 0.

Example

- The probability of flipping a fair coin and getting tails is 0.50, or 50%. If a coin is flipped 10 times, there is no guarantee, that exactly 5 tails will be observed, the proportion of tails can range from 0 to 1.

a) Random experiment

In the study of probability, any process of observations is referred to as an experiment.

- A **random experiment** is a situation involving chance or probability that leads to well-defined results called outcomes.
- An experiment is called **random experiment** if its outcomes cannot be predicted.
- Then, the results of an observation are called outcomes of the experiment.
- So an outcome is the result of a single trial of an experiment (H: head; T: tail)

Examples

Example

To throw a die , to roll a coin , to draw a card from a deck (set of cards), or selecting a diamond , a spade , a club , or a heart [Throwing an ace, a king, a jack, a queen of diamonds, spades, clubs, or hearts.

b) Sample Space

- The set of all possible outcomes of a random experiment is called the Sample Space (**universal set**) and it is denoted by S.
- An element in is called a **sample point.**
- An **event** is one or more outcomes of a probability experiment.
- **Complement of an event:** all the elements in the sample space except the given elements.
- **Mutually exclusive events or Disjoint events** : two events which cannot happen at the same time.

Examples

Example of rolling a die

- A single 6-sided die is rolled. The possible outcomes of this experiment are 1, 2,3,4,5, and 6.
- $\Omega = U = S$ is the set of all possible outcomes.
- A single outcome of this experiment is rolling a 1, a 2, a 3,...
- Rolling an even number (2, 4, or 6) is an event.
- Rolling an odd number (1, 3, or 5) is also an event.

Examples of random experiments:

- The roll (throw) of a die: $\Omega = U = S = \{1, 2, 3, 4, 5, 6\}$.
- The toss of a coin: $\Omega = U = S = \{H, T\}$.

Exercise

Exercise 1:

Find the sample space for the experiment of tossing a coin

1. Once
2. Twice

Solution

1. There are two possible outcomes, head or tail. Thus $S = \{H, T\}$
2. There are four possible outcomes : pairs of heads and tails: $S = \{HH, HT, TH, TT\}$

Tree Diagrams

- Tree diagram is a graphical way of listing all the possible outcomes.
- The outcomes are listed in an orderly fashion, so listing all of the possible outcomes is easier than just trying to make sure that you have them all listed.
- It is called a tree diagram because of the way it looks.
- The first event appears on the left, and then each sequential event is represented as branches off of the first event.
- The tree diagram to the right would show the possible ways of flipping two coins.
- The final outcomes are obtained by following each branch to its conclusion:
- They are from top to bottom: $\{HH, HT, TH, TT\}$

Exercise 2:

1. Find the sample space for the experiment of roll a die

    1. Once
    2. Twice

Solution

1. $S = \{1, 2, 3, 4, 5, 6\} \Rightarrow \#S = 6$
2. There are 36 possible outcomes:$S = \{1, 2, 3, 4, 5, 6\} * \{1, 2, 3, 4, 5, 6\} \Rightarrow \#S = 36$

	1	2	3	4	5	6
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

Probability

Measuring probabilities using a calibration experiment

- Probabilities are generally hard to measure. It is easy to measure probabilities of events that are extremely rare or events that are extremely likely to occur.
- But consider your probability for the event "There will be a white Christmas this year".
- You can remember years in the past where there was snow on the ground on Christmas.
- Also you can recall past years with no snow on the ground.
- So the probability of this event is greater than 0 and less than 1. But how do you obtain the exact probability?

Definition

If the possibility space S has a finite number of sample points, then, we denote this number by $n(s)$. Consider an event A which is a subset of S, then $n(A) \leq n(S)$.

Tossing a die. **Outcomes:**

- The possible outcomes of this experiment is $S = \{1, 2, 3, 4, 5, 6\}$.
- Let A_1 be the event of obtaining an even number: $A_1 = \{2, 4, 6\}, \quad \Rightarrow \#A_1 = 3$.
- Let A_2 be the event of obtaining an odd number :$A_2 = \{1, 3, 5\}, \quad \#A_2 = 3$.

Therefore, the probability is the measure of how likely an event is.

Exercise 3

Tossing a die

1. Let A be the event that the appeared number is even. $A = \{2, 4, 6\}$.
2. Let B be the event that the occurring number is odd. $B = \{1, 3, 5\}$.
3. Let C be the event that the occurring number is less than 4. $C = \{1, 2, 3\}$.

Questions

1. What is the probability of each outcome?
2. What is the probability of rolling an even number? An odd number?

Probabilities:

The probability of an event A_1, written $p(A_1)$ is given by:

$$p(A_1) = \frac{n(A_1)}{n(S)} = \frac{number\ of\ possible\ outcomes\ of\ event\ A_1}{number\ of\ total\ outcomes} = \frac{the\ number\ of\ ways\ event\ A\ can\ occur}{the\ total\ number\ of\ possible\ outcomes}$$

$$p(A_1) = \frac{3}{6} = \frac{1}{2}$$

$$p(A_2) = \frac{3}{6} = \frac{1}{2}$$

If $p(A) > p(B)$, then A is more likely to occur than event B.

If $p(A) = p(B)$, then events A and B are equally likely to occur.

- The probability of each outcome is always the same $\frac{1}{6}$. The outcomes are equally likely to occur.
  Equally likely events: events which have the same probability of occurring.

$$\begin{aligned} P(1) &= \frac{Numberofwaystorolla1}{Totalnumberofsides} = \frac{1}{6} \\ P(2) &= \frac{1}{6}, \ldots \end{aligned}$$

- Classical probability uses the sample space to determine the numerical probability that an event will happen. It is also called theoretical probability.

Examples

1. 1. A box contains 6 red balls, 5 green, 8 blue and 3 yellow. A single ball is chosen at random from the box. What is the probability of choosing
   - a red ball?
   - a green ball?
   - a yellow ball?
   - a red or green ball?
2. The outcomes are not equally likely to occur. You are more likely to choose a blue ball than any other color. You are least likely to choose a yellow ball.
3. Choose a number at random from 1 to 5. What is the probability of each outcome? What is the probability that the number chosen
   - is even?
   - is odd ?
   - is prime?

Axioms of Probability

1. $P(S) = 1$
2. $A \subset S$, $0 \leq P(A) \leq 1$. All probabilities are between 0 and 1 inclusive.
3. $A, B \subset S \Rightarrow P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
4. If $A \cap B = \emptyset$. A and B are mutually exclusive events in S. Then $P(A \cup B) = P(A) + P(B)$.
5. The probability of an event which cannot occur (impossible event) is 0.
6. The probability of an event which is not in the sample space is zero.
7. The probability of an event which must occur is 1.
8. If A' is the event of not occurring the event, then $P(A') = 1 - P(A)$.

Definition

1. If two events A and B are such that $A \cup B = S$, then $P(A \cup B) = 1$. Hence, A and B are called **Exhaustive events**.
2. If $A \cap B = \emptyset$. A and B are mutually exclusive events in S. Then $P(A \cup B) = P(A) + P(B)$.
3. **Independent event:** We say that two events are independent if the occurrence of one event does not affect the occurrence of an other.
4. **Dependent events:** If the occurrence of one event affects the occurrence of the the other, then two events are called dependent events.

(iii) Conditional probability

The conditional probability $P(A/B)$ is defined as the probability of an event A occurring given that B has already occurred.

$$P(A|B) = \frac{P(A \cap B)}{P(B)},\ P(B) \neq 0 \Rightarrow P(A \cap B) = P(A/B) * P(B)$$

$$P(B|A) = \frac{P(A \cap b)}{P(A)},\ p(A) \neq 0, \Rightarrow P(A \cap B) = P(B/A) * P(A)$$

Example 1:

Gender \ Smoke	Yes	No	Total
Male	19	41	**60**
Female	12	28	**40**
Total	**31**	**69**	**100**

1. What is the probability of a randomly selected individual being a male who smokes?
2. What is the probability of a randomly selected individual being a male?
3. What is the probability of a randomly selected individual smoking?
4. What is the probability that the male smokes?
5. What is the probability that a randomly selected smoker is male?

Solution

1. The number of Male and Smoke divided by the total is $19/100$.
2. This is the total for male divided by the total is $60/100$. Since no mention is made of smoking or not smoking, it includes all the cases.
3. Again, since no mention is made of gender, this is a marginal probability, the total who smoke divided by the total is $31/100$.
4. Well, 19 males smoke out of 60 males, so $19/60$.
5. This time, you're told that you have a smoker and asked to find the probability that the smoker is also male. There are 19 male smokers out of 31 total smokers, so is $19/31$.

Example 2:

1. Given a heart is picked at random from a park of playing cards, find the probability of that it is a picture card.

Solution:

(iv) Independent events

- If either A or B can occur without being affected by the other, then the 2 events are said to be independent and $P(A \cap B) = P(A) \times P(B)$.
- **Dependent events:** two events A and B are dependent if the first event affects the outcome or occurrence of the second event in a way the probability is changed.
- If A and B are independent events, then

$$
\begin{aligned}
P(A|B) &= \frac{P(A \cap B)}{P(B)} = \frac{P(A) * P(B)}{P(B)} = P(A) \\
P(B|A) &= \frac{P(A \cap b)}{P(A)} = \frac{P(A) * P(B)}{P(A)} = P(B)
\end{aligned}
$$

Exercises

1. Events A and B are such that $P(A) = \frac{1}{3}$, $P(A \cap B) = \frac{1}{12}$. If A and B are independent, find
   a) $P(B)$
   b) $P(A \cup B)$
2. Events A and B are such that $P(A) = \frac{1}{4}$, $P(A|B) = \frac{1}{2}$ and $P(B|A) = \frac{2}{3}$.
   a) Are A and B independent?
   b) Are A and B mutually exclusive events?
   c) Find $P(A \cap B)$ d) Find $P(B)$

(v) Bayes' theorem

A formula which allows one to find the probability that an event occurred as the result of a particular previous event.
Suppose that $A_i\,(i = 1, 2, 3, \ldots, n)$ are mutually exclusive and exhaustive events so that $U_{i=1}^{n} A_i = S$, the possible space and B an arbitrary event on s.

$$P(A_i|B) = \frac{P(B|A_i) * P(A_i)}{\sum_{i=1}^{n} P(B|A_i) * P(A_i)} \quad for\ i = 1, 2, 3, \ldots, n$$

Exercise

Three machines A, B and C produce $50\%, 30\%$ and 20% of the total number of materials made respectively in a factory. The percentages of defective materials of these machines are respectively $3\%, 4\%$ *and* 5%.

If one takes a material at random.

a) What is the probability so that this material is defective?

b) Calculate the probability so that this material has been produced by the machine A?

Basic Definitions and Rules of probability

- An **experiment** is defined as any planned process of data collection. For an experiment we define an **event** to be any collection of possible outcomes.
- A **conditional probability** is the probability of one event given that another event has occurred. A **simple event** is an event that consists of exactly one outcome.
- In Probability "OR" means the union that is either can occur and in probability "AND" means intersection that is both must occur.
- Two events are **mutually exclusive** if they cannot occur simultaneously.

Characteristics of probability

1. P(E) is always between 0 and 1.
2. The sum of the probabilities of all simple events must be 1.
3. P(E) + P (not E) = 1
4. $P(\emptyset) = 0$
5. If $A \subseteq B$, then $P(A) \leq P(B)$
6. If A and B are mutually exclusive then $P(A \cup B) = P(A) + P(B)$
7. If A and B are not mutually exclusive then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ (Additive law)
8. If A and B are independent then $P(A \cap B) = P(A)P(B)$ (multiplicative law)
9. For conditional probability $P(A|B) = \dfrac{P(A \cap B)}{P(B)}$; $P(B|A) = \dfrac{P(B \cap A)}{P(A)}$

The multiplication rule for probabilities when events are not independent can be used to derive one form of an important formula called Baye's theorem. Since $P(A \cap B)$ equals both $P(A|B) \times P(B)$ and $P(B|A) \times P(A)$, these latter two expressions are equal. Assuming that $P(B)$ and $P(A)$ are not equal to zero, we can solve for one in terms of the other as follows:

$P(A|B) \times P(B) = P(B|A) \times P(A)$ Then;

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

$$P(B|A) = \frac{P(A|B) \times P(B)}{P(A)}$$

Note that, it is an extension of the multiplicative rule.

If $A \subseteq B$, then $A \cap B = A$, so $P(A|B) = \dfrac{P(A)}{P(B)}$

Example 1

A die is rolled and the number showing recorded. Given that the number rolled was even, what is the probability that it was a six?

Solution

Let E denote the event "even" and F denote the event "a six". Clearly $F \subseteq E$, so

$$\begin{aligned} P(F|E) &= \frac{P(F)}{P(E)} \\ &= \frac{1/6}{1/2} \\ &= 0.3 \end{aligned}$$

Example 2

Let's consider the following hypothetical data set on employment

Table: hypothetical data set on employment

Race	Population	Currently Employed
Black	25234324	13423435
White	5843853	4952423
Other	8423543	6443543
Total	39501720	24819401

Solution

Let A_1 be an event that an individual is black, A_2 that he/she is white and A_3 is that he/she is in the other group.

$$P(A_1) = \frac{25234324}{39501720} = 0.638815829;$$
$$P(A_2) = \frac{5843853}{39501720} = 0.147939204;$$
$$P(A_3) = \frac{8423543}{39501720} = 0.213244968.$$

An individual from this population can belong to one and only one group, so events A_1, A_2, and A_3 are mutually exclusive, thus we can use the additive law so that:

$$\begin{aligned} P(A_1 \cup A_2 \cup A_3) &= P(A_1) + P(A_2) + P(A_3) \\ &= 0.638815829 + 0.147939204 + 0.213244968 \end{aligned}$$

Now let B be the event that an individual is employed, then

$$\begin{aligned} P(B) &= 24819401/39501720 \\ &= 0.628311906 \end{aligned}$$

If we look at each race group separately, then;

- The probability that an employed individual is black $P(B_1) = \dfrac{13423435}{25234324} = 0.5320$ or $P(B/A_1) = \dfrac{13423435/39501720}{25234324/39501720} = .5320$
- The probability that an employed individual is white $P(B_2) = 4952423/5843853 = 0.8475$
- The probability that an employed individual is other $P(B_3) = 6443543/8423543 = 0.7649$

We can express the event B (someone is employed) as the union of three mutually exclusive events, $A_1 \cap B$ (the event that an individual is black and is employed), $A_2 \cap B$ (the event that one is white and is employed), $A_3 \cap B$ (the event that one is other and employed). That is:
$B = (A_1 \cap B) \cup (A_2 \cap B) \cup (A_3 \cap B)$.
Since everyone who is employed can be place in to one and only one of these three categories, we can thus apply the additive rule, That is:

$$\begin{aligned} P(B) &= P\left[(A_1 \cap B) \cup (A_2 \cap B) \cup (A_3 \cap B)\right] \\ &= P(A_1 \cap B) + P(A_2 \cap B) + P(A_3 \cap B). \end{aligned} \tag{1}$$

And applying the multiplicative rule to each term separately we get;

$$\begin{aligned} P(B) &= P(A_1)P(B/A_1) + P(A_2)P(B/A_2) + P(A3)P(B/A_3) \\ &= (0.6388)(0.5320) + (0.1479)(0.8475) + (0.2132)(0.7649) \\ &= 0.6283 \end{aligned}$$

This is the same value we got originally as the probability that an individual is employed.
Note that the equation in (1) can be used when we are unable to calculate $P(B)$ directly.

Suppose now we want to calculate $P(A_1/B)$, that is the probability that someone is black given that he/she is employed. The multiple rule gives:

$$P(A_1 \cap B) = P(B)P(A_1/B);$$

$$P(A_1/B) = \frac{P(A_1 \cap B)}{P(B)}; \quad \text{by dividing boh sides by} \quad P(B)$$

Applying the multiplicative rule to the numerator of the right hand sides we get;

$$P(A_1/B) = \frac{P(A_1)P(B/A_1)}{P(B)};$$

Or $\quad P(B) = P(A_1)P(B/A_1) + P(A_2)P(B/A_2) + P(A3)P(B/A_3)$, so;

$$P(A_1/B) = \frac{P(A_1)P(B/A_1)}{P(A_1)P(B/A_1) + P(A_2)P(B/A_2) + P(A3)P(B/A_3)}. \tag{2}$$

The formula in equation (2) **represents Bayes' theorem**.

Substituting the numerical values of all the terms in equation (2) we get;

$$P(A_1/B) = \frac{(0.6388)(0.5320)}{(0.6388)(0.5320) + (0.1479)(0.8475) + (0.2132)(0.7649)}$$
$$= 0.5409$$

So the probability that one is black given that he is employed is 0.5409, which we can get directly from the data. Among 24819401 employed people there are 13423435 black, thus
$P(A_1/B) = 13423435/24819410 = 0.5408$

Some Definitions

A **Variable** is a characteristic that can be measured or categorized.
A **random variable** is when values of a variable occur by chance and then the variable is called a random variable.
Discrete random variable can only take finite or countable values. Eg. Sex, parity, race etc
Continuous random variable can take any value within a specified interval. Eg. Blood pressure, weight etc
Probability distribution – all random variables has a corresponding probability distribution, which uses the theory of probability to describe the behaviour of a random variable.

Example

Consider the data below which show the frequency distribution of the number of babies that women have had. Here let X be the discrete random variable representing the number of children a woman have had, if the study child was the first child then X=1, if he/she the second then X=2, and so on. The table 2 shows the probability distribution of X, i.e. P(X), which is the proportion of a woman with 1,2, ... children (frequency distribution). We observe all possible outcomes of X, so the probabilities add up to 1 (exhaustive trial)

Table: Table of number of children

Parity	Count	Percentage
1	6076	43.06
2	4267	30.24
3	1977	14.01
4	955	6.77
5	479	3.39
6	216	1.53
7	86	0.61
8	31	0.22
9	13	0.09
10	9	0.06
11	1	0.01

Suppose we want to know the probability that a particular mother's child is her 3rd child, from the table P $(X=3)=0.14$ What is the probability that a chosen mother's child is her 4^{th} or 5^{th}?

Note

If the frequency groups or categories become many it will be difficult to use frequency distribution as in the above example. Instead one appeals to known theoretical probability distributions. Most measurements in real life take the form of known theoretical distributions. **Example**: Weight and Age have roughly normal distributions

Exercise

1. A die is thrown twice and the number on each throw is recorded. What is the probability that at least one six occurs?

2. If two events, A and B, are such that $P(A) = 0.5$, $P(B) = 0.3$, and $P(A \cap B) = 0.1$, find the following:

1. $P(A|B)$
2. $P(B|A)$
3. $P(A|A \cap B)$
4. $P(A|A \cup B)$
5. $P(A \cap B|A \cup B)$

Quiz I

1. Two fair coins are tossed. Given that at least one head appears, what is the probability that both coins show heads?
2. In how many ways a committee consisting of 3 men and 2 women, can be chosen from 5 men and 4 women?
3. Let A and B be possible events of an experiment such that $P(A) = 0.4$, $P(B) = p$, $P(A \cap B) = 0.3$. Find the value of p such that A and B are independent events.
4. If two events, A and B, are such that $P(A) = 0.5$, $P(B) = 0.4$, and $P(A \cap B) = 0.2$, find the following:
    1. $P(A')$.
    2. $P\left[(A \cap B)|(A \cup B)\right]$.

Probability Density function

Def

- In probability theory, a probability density function (pdf), or density of a continuous random variable is a function that describes the relative likelihood for this random variable to occur at a given point.
- The probability for the random variable to fall within a particular region is given by the integral of this variable's density over the region.
- The probability density function is non negative everywhere, and its integral over the entire space is equal to one. That is for absolutely continuous uni-variate distribution

Some probability distribution of random variables (r.v)

- **Random variable** is the measurements of a random variable vary in a seemingly random and unpredictable manner.
- A random variable assumes a unique numerical value for each of the outcomes in the sample space of the probability experiment.
- If we toss a coin twice, the number of Heads obtained could be 0, 1 or 2.
- **Random Variable** is a variable whose values are determined by chance.

Example

Consider the discrete random variable from the experiment of tossing two fair dice and given by the sum of points. We have

$$X(a, b) = a + b; X(\omega(= \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

. $X(a, b)$ can be described by the following table:

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

The probability function is given by

x_i	2	3	4	5	6	7	8	9	10	11	12
$f(x_i)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

The cumulative function is given by

x_i	2	3	4	5	6	7	8	9	10	11	12
$F(x_i)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{6}{36}$	$\frac{10}{36}$	$\frac{15}{36}$	$\frac{21}{36}$	$\frac{26}{36}$	$\frac{30}{36}$	$\frac{33}{36}$	$\frac{35}{36}$	$\frac{36}{36}$

The probabilities of these occurrences are:

P (No heads) =P (TT) = (1/2)* (1/2) =1/4

P (1 head) =P (TH) +P (HT) = (0.5*0.5) + (0.5*0.5) =0.5

P (2 heads) =P (HH) =0.5*0.5=0.25

The probability distribution is often written as follow:

X	0	1	2
P(X=x)	0.25	0.5	0.25

Discrete random variables

Let X be a variable which can take only values $x_1, x_2, x_3, \ldots, x_n$ .The probability associated with these values are $P_1, P_2, \ldots, P_n$. Where

$$P(X = x_1) = P_1$$
$$P(X = x_2) = P_2$$
$$\ldots\ldots\ldots\ldots\ldots\ldots$$
$$\ldots\ldots\ldots\ldots\ldots\ldots$$
$$P(X = x_n) = P_n$$

Then X is a **discrete random variable** if $\sum_{i=1}^{n} P_i = 1$ or $\sum_{all\ x} P(X = x) = 1$

We always denote a random variable by a capital letter X,Y,Z....and the particular values by lower case letter x,y,z....

Probability density function (pdf)

This is the function that is responsible for allocating probabilities. It is written as

$f(x_i) = P(X = x_i) \quad with \quad 1 \leq i \leq n$.

x_1	$f(x_1) = P_1$
x_2	$f(x_2) = P_2$
x_3	$f(x_3) = P_3$
……	…….
x_n	$f(x_n) = P_n$

$0 \leq f(x_i) \leq 1 \text{ and } \sum_{i=1}^{n} f(x_i) = 1$

Example: The following table gives the pdf of X

i	1	2	3	4
x_i	-3	-2	1	4
$P(X = x_i)$	0.1	0.2	0.3	0.4

Characteristics of a random variable.

Expectation:

Let X be a r.v with pdf f(x)=P(X=x) and x is discrete. The expectation of X written $E(X)$ is given by

$$E(X) = \sum x\,P(X = x) \;\; or \;\; E(X) = \sum_{i=1}^{n} x_i\,P_i \;\; or \;\; E(X) = \sum_{i=1}^{n} x_i f(x_i)$$

It is the average of the numbers giving the gains and the losses weighted by their probabilities.

The game permits to expect an average gain or an average loss (by part) of E(X).

One says that the game is favorable if E(X) is positive and it is not favorable if E(X) is negative.

If E(X) =0, the game is balanced.

Theorems:

1. Let consider an r.v X and a real number k:

$$E(kX) = k\,E(X)$$
$$E(X+k) = E(X) + k$$

2. Let consider X and Y, two r.v:

$$E(X+Y) = E(X) + E(Y)$$

Example: A random variable has pdf as shown. Find E(X).

X	-2	-1	0	1	2
P(X=x)	0.3	0.1	0.15	0.4	0.05

Solution

$$E(X) = \sum_{all\ x} x\,P(X = x) = (-2 * 0.3) + (-1 * 0.1) + (0 * 0.15) + (1 * 0.4) + (2 * 0.05) = -0.2$$

The expectation of any function of X, E[g(x)]

Let X be a r.v and pdf f(x) =P(X=x). Let also g(x) be a function of the r.v. X. Then , the expected value of g(x) written E[g(x)] is given by: $E[g(x)] = \sum_{all\ x} g(x)\, P(X = x)$

Example: The r.v X has pdf P(X=x) for x=1, 2, 3.

X	1	2	3
P(X=x)	0.1	0.6	0.3

Compute : a) E(3) b) E(X) c) E(5X) d) E(5X+3)

Solution

a) $E(X) = \sum_{all\ x} x P(X = x)$

$E(x) = \sum_{all\ x} 3P(X = x) = 3\sum_{all\ x} P(X = x) = 3*(0.1+0.6+0.3) = 3*1 = 3$

b) $E(X) = \sum_{all\ x} x P(X = x) = (1*0.1) + (2*0.6) + (3*0.3) = 2.2$

c) $E(5X) = \sum_{all\ x} 5x P(X = x) = (5*1*0.1) + (5*2*0.6) + (3*5*0.3) = 11$

d) $E(5X+3) = \sum_{all\ x} (5x+3) P(X = x) = 8*0.1 + 13*0.6 + 18*0.3 = 14$

Variance (Var X) and standard deviation σ_x

Let X be a r.v having the pdf $f(x) = P(X = x), x \in IR$ and the $E(X) = \mu$ where μ is constant. The variance of X, written Var (X) is given by: $Var\ X = E[(X-\mu)^2]$ .Alternatively,

$$
\begin{aligned}
Var\ X &= E\left[(X-\mu)^2\right] \\
&= E\left(X^2 - 2\mu X + \mu^2\right) \\
&= E\left(X^2\right) - 2\mu * E(X) + E\left(\mu^2\right) \\
&= E\left(X^2\right) - 2 * \mu * \mu + \mu^2 \\
&= E\left(X^2\right) - 2\mu^2 + \mu^2 \\
&= E\left(X^2\right) - \mu^2
\end{aligned}
$$

$$
\begin{aligned}
Var\ \ X &= E\left(X^2\right) - [E(X)]^2 \\
&= \sum_{i=1}^{n} x_i^2 * P_i - E^2(X)
\end{aligned}
$$

$$
\sigma_X = \sqrt{Var\ X}
$$

Properties

$$Var(k) = 0$$

$$Var(kX) = k^2 Var(X)$$

$$Var(X + k) = Var(X)$$

$$Var(kX + +b) = k^2 Var(X)$$

$$\sigma(kX) = \sigma_{kX} = |k| * \sigma_X$$

Example:

An r.v X has the following distribution:

X	1	2	3	4	5
P(X=x)	0.1	0.3	0.2	0.3	0.1

Find E(X), Var(X) and σ_x

Solution

a) $E(X) = \sum_{all\ x} xP(X = x) = (1*0.1) + (2*0.3) + (3*0.2) + (4*0.3) + (5*0.1) = 3$

b) $Var\ X = E(X^2) - [E(X)]^2$

$E(X^2) = \sum_{all\ x} x^2 P(X = x) = (1*0.1) + (4*0.3) + (9*0.2) + (16*0.3) + (25*0.1) = 10.4$

$So\ that\ Var\ X = 10.4 - 3^2 = 1.4$

$\sigma_x = \sqrt{1.4} = 1.1832$

Mathematical Expectation and Variance

Expected value

Let X be a random variable with probability distribution $f(x)$.

1. If X is discrete, then the **mean** (μ), or **expected value** $(E(X))$, of X is given by; $\mu = E(X) = \sum_x xf(x)$.

Properties

For a and b constants and X and Y random variables.

1. $E[b] = b$;
2. $E[aX] = aE[X]$;
3. $E[aX + b] = aE[X] + b$;
4. $E[X + Y] = E[X] + E[Y]$;
5. $E[XY] = E[X]E[Y]$; for X and Y independent.

Variance

The variance of a random variable X is given by

$$\sigma^2 = E\left[(X-\mu)^2\right] = E(X^2) - E^2(X)$$

Exercise 1

Let the random variable X represent the number of defective parts for a machine when 3 parts are sampled from a production line and tested. The following is the probability distribution of X. Calculate the variance σ^2.

x	0	1	2	3
$f(x)$	0.51	0.38	0.10	0.01

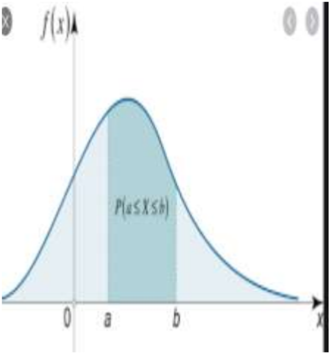
Continuous random variable

The probability density function is non negative everywhere, and its integral over the entire space is equal to one. That is for absolutely continuous uni-variate distribution.

So

1. $f(x) \geq 0$, for all $x \in \mathbb{R}$.
2. $\int_{\infty}^{\infty} f(x)\mathrm{d}x = 1$.
3. $P(a < X < b) = \int_{a}^{b} f(x)\mathrm{d}x$

Definition. The function $f(x)$ is a **probability density function** for the c.d.v X, defined over $\Re$, if: i. $f(x) \geq 0, \forall x \in \Re$ ii. $\int_{-\infty}^{+\infty} f(x)dx = 1$ iii. $p(a < X < b) = \int_a^b f(x)dx$



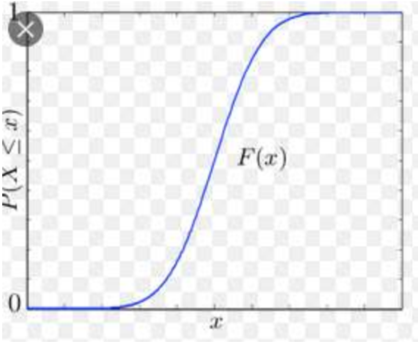


Fig 3.4. $p(a < X < b) = \int_a^b f(x)dx$ **Fig3.5. Cumulative distribution function**

Characteristics of Continuous random variable

Expectation, variance and standard deviation

1. The expectation value, $\mu = E(X) = \int_{-\infty}^{\infty} xf(x)\,dx$.
2. The variance, $\sigma^2 = Var(X) = E(X^2) - (\mu)^2 = \int_{-\infty}^{\infty} x^2 f(x)\,dx - \left(\int_{-\infty}^{\infty} xf(x)\,dx\right)^2$.
3. Standard deviation, $Stev(X) = \sqrt{Var(x)}$.

Example

Suppose that the error in the reaction temperature, in C, for a controlled laboratory experiment is a continuous random variable X having the probability density function

$$f(x) = \begin{cases} \dfrac{x^2}{3} & -1 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

(a) Verify that $f(x)$ is a density function.

(b) Find $P(0 < X \leq 1)$.

(c) Determine the expected value

(d) Determine the variance and standard deviation.

(e) Determine the standard deviation.

Solution

(a) Obviously, $f(x) \geq 0$. $\int_{\infty}^{\infty} f(x)\mathrm{d}x = \int_{-1}^{2} \frac{x^2}{3}\mathrm{d}x = \frac{x^3}{9}\Big|_{-1}^{2} = 1$

(b) $P(0 < X \leq 1) = \int_{0}^{1} \frac{x^2}{3}\mathrm{d}x = \frac{x^3}{9}\Big|_{0}^{1} = \frac{1}{9}$

(c) $E(X) = \mu = \int_{-\infty}^{\infty} xf(x)\, dx = \int_{-1}^{2} x\frac{x^2}{3}\, dx = \int_{-1}^{2} \frac{x^3}{3}\, dx = \frac{15}{12}$.

(d) $Var(X) = E(X^2) - (\mu)^2 = \int_{-1}^{2} x^2\frac{x^2}{3}\, dx - \left(\frac{15}{12}\right)^2 =$
$\int_{-1}^{2} \frac{x^4}{3}\, dx - \left(\frac{15}{12}\right)^2 = 0.6375$.

(e) $\sigma = \sqrt{Var(X)} = 0.8984$.

Exercise 2

The weekly demand for a drinking-water product, in thousands of litres, from a local chain of efficiency stores is a continuous random variable X having the probability density

$$f(x) = \begin{cases} 2(x-1), & 1 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$$

Calculate the expected value (mean) and variance.

Some Probability Distributions

Discrete Probability distributions

1. Discrete uniform distribution.
2. **Binomial Distributions.**
3. Hyper geometric distribution.
4. **Poisson Distribution.**

Continuous Probability Distributions

1. Continuous Uniform Distribution.
2. Exponential distribution.
3. **Normal distribution**

Discrete uniform distribution

- The simplest of all discrete probability distributions is one where the random variable assumes each of its values with an equal probability. Such a probability distribution is called a **discrete uniform distribution.**

Definition

If the random variable X assumes the values $1, 2, \ldots, n$, with equal probabilities, then the discrete uniform distribution is given by

$$f(X_i) = \frac{1}{n}, \quad X_i \in \{1, 2, \ldots, n\}.$$

- The moment generating function of the binomial distribution is given by

$$G_X(t) = E(e^{tX}) = \frac{e^t(1 - e^{nt})}{n(1 - e^t)}.$$

cont...

Proposition

The **mean** and **variance** of the discrete uniform distribution are given

$$\begin{aligned} f(X_i) &= \frac{1}{n}, \quad X_i = 1, 2, \ldots, 6. \\ E(X) &= \frac{1}{n}\sum_{i=1}^{n} X_i = \frac{n+1}{2} \\ \sigma_x^2 &= \frac{1}{n}\sum_{i=1}^{n}(X_i - E(x))^2 = E(x^2) - E^2(X) = \frac{n^2-1}{12}. \end{aligned}$$

Example

When a fair die is tossed, each element of the sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$ occurs with probability $\frac{1}{6}$.

Example with answer

Example

When a fair die is tossed, each element of the sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$ occurs with probability $\frac{1}{6}$.

Therefore, we have the uniform distribution, with

$$
\begin{aligned}
E(X) = \mu &= \frac{1}{n}\sum_{i=1}^{n} X_i = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5; \\
Variance = \sigma_x^2 &= \frac{1}{n}\sum_{i=1}^{n}(X_i - E(x))^2 = E(x^2) - E^2(X) = 2.92.
\end{aligned}
$$

Special discrete probability distributions

Binomial distribution

- **Binomial experiment** is an experiment with a fixed number of independent trials.
- Each trial can only have two outcomes, or outcomes which can be reduced to two outcomes.
- The probability of each outcome must remain constant from trial to trial.
- **Binomial distribution** is the distribution of the outcomes of a binomial experiment with their corresponding probabilities.

Binomial distribution

- Let consider an experiment with n **independent trials and only two possible outcomes**.
- We call one of the outcomes **successful outcome** and the probability of its occurring is $P(s) = p$.
- The other outcome is called **fail outcome** and the probability of its occurring is $P(f) = q = 1 - p$.

The probability for getting exactly k successes in n trials is

$$P_k = B(k, n, p) = B(n, p) = \binom{n}{k} p^k q^{n-k} \text{ and } k = 0,1,2,\ldots,n.$$

Then the pdf of X is given by $f(x) = P(X = x) = \binom{n}{k} p^k q^{n-k} \text{ and } k = 0,1,2,\ldots,n$

We write $X \rightarrow Bin(n, p)$ and read "X is a random variable following the binomial distribution with parameter n and p".

Example: The probability that a person supports party A is 0.6.Find the probability that a randomly selected sample of 8 voters, there are:

a) Exactly 3 who support party A.

b) More than 5 who support party A

Solution: Consider "support of party A to be success". Then p=0.6 and q=1-0.6=0.4.

Let X be the r.v." the number of party A supporters"

$X \sim Bin(n,\ p)$

$X \sim Bin(8,\ 0.6)$ and its pdf is given by $f(x) = P(X = x) = \binom{n}{k} p^k q^{n-k}$ $and\ k = 0,1,2,\ldots,n$

Thus $f(x) = P(X = x) = \binom{8}{k} (0.6)^k (0.4)^{8-k}$ $and\ k = 0,1,2,\ldots,8$

$$f(x) = P(X = x) = \binom{8}{k}(0.6)^k (0.4)^{8-k} \quad and \quad k = 0,1,2,\ldots,8$$

Thus

a) k=3

$$\mathbf{P(X=3)}=\binom{8}{3}(0.6)^3(0.4)^{8-3} = 0.124$$

P(X>5)=P(X=6)+P(X=7)+P(X=8)

$$=\binom{8}{6}(0.6)^6(0.4)^{8-6}+\binom{8}{7}(0.6)^7(0.4)^{8-7}+\binom{8}{8}(0.6)^8(0.4)^{8-8}$$

$$=\binom{8}{6}(0.6)^6(0.4)^2+\binom{8}{7}(0.6)^7(0.4)^1+\binom{8}{8}(0.6)^8(0.4)^0$$

$$\mathbf{=0.315}$$

Expectation value and variance.

If the r.v X is such that $X \sim Bin(n,\ p)$, then $E(X) = np \ and \ Var \ X = npq \ where \ \ q = 1 - p$

Example: If the probability that it will be a fine day is 0.4. Find the expected number of fine days in a week end and the standard deviation.

Solution

Let "fine day" be "success". Then, p=0.4 and q=0.6.

Let X be an r.v. "number of fine days in a week ".Then $X \sim Bin(n,\ p)$, where n=7 and p=0.4.

Hence $X \sim Bin(7,\ 0.4)$

a) $E(X) = np = 7*(0.4) = 2.8$

b) $Var\ X = npq = 7*(0.4)*(0.6) = 1.68$

$\Rightarrow \sigma = \sqrt{Var\ X} = \sqrt{1.68} = 1.2961$

Mode

Let the r.v. X be such that $X\ Bin(n,\ p)$. To compute the mode of the probability distribution of X, we only consider the values of X close to E(X).

Example:

If $X \sim Bin(10,\ 0.45)$, find the mode of the probability distribution of X.

Solution: Here X has pdf as follow: $f(x) = P(X = x) = \binom{n}{k} p^k q^{n-k}$ *and* $k = 0, 1, 2, \ldots, n$

So that $P(X = x) = \binom{10}{k} (0.45)^k (0.55)^{n-k}$ *and* $k = 0, 1, 2, \ldots, 10$

But E(x) =np=10*0.45=4.5

Now $P(X = 3) = \binom{10}{3} (0.45)^3 (0.55)^{10-3} = 0.1664$

$P(X = 4) = \binom{10}{4} (0.45)^4 (0.55)^{10-4} = 0.2383$

$P(X = 5) = \binom{10}{5} (0.45)^5 (0.55)^{10-5} = 0.2340$

$P(X = 6) = \binom{10}{6} (0.45)^6 (0.55)^{10-6} = 0.1595$

Here X with the highest probability is 4. Therefore, the mode of X is 4.

Exercises

1. We are interested in the probability of tossing exactly 7 heads in 10 tosses.
2. A real estate agent has 5 contacts and believes that for each contact, the probability of making a sale is 0.40. What is the probability of making exactly 2 sales?
3. Find the probability of getting between 3 and 6 heads inclusive in 10 tosses of a fair coin by using the binomial distribution.
4. A student takes a ten-question true/false exam.
   - Find the probability that the student gets exactly six of the questions right simply by guessing the answer on every question.
   - Find the probability that the student will obtain a passing grade of 60% or greater simply by guessing.

Poisson distribution

- **Poisson distribution** is a probability distribution used when a density of items is distributed over a period of time.
- **The sample size needs to be large and the probability of success to be small.**
- Let X be a discrete r.v. It is said to follow the Poisson distribution if its pdf is of the form:

$P(X=x)=\dfrac{e^{-\lambda}\lambda^{x}}{x!}$ *for* $x=0,1,2,\ldots\ldots$ *to* inf *inity* *where* λ *is the parameter of the*

We write : $X \sim \mathrm{P}_0(\lambda)$

Then the Poisson distribution is defined by

$p(k,\lambda)=p_0(\lambda)=\dfrac{\lambda^{k}e^{-\lambda}}{k!}$, $k=0,1,2,\ldots\ldots$ *where* λ *is anycons* tan *t*.

This distribution is an infinitely countable distribution.

Solution

$$P(X=x)=\frac{e^{-3.5}(3.5)^x}{x!}, \quad x=0,1,2,\ldots\ldots$$

a) $P(X=4)=\dfrac{e^{-3.5}(3.5)^4}{4!}=0.1888\ldots..$

b)

$$\begin{aligned}P(X\le 2)&=P(X=0)+P(X=1)+P(X=2)\\&=\frac{e^{-3.5}(3.5)^0}{0!}+\frac{e^{-3.5}(3.5)^1}{1!}+\frac{e^{-3.5}(3.5)^2}{2!}\\&=e^{-3.5}\left(1+3.5+\frac{(3.5)^2}{2!}\right)=0.3208\end{aligned}$$

c)

$$\begin{aligned}P(X>1)&=1-P(X\le 1)\\&=1-[P(X=0)+P(X=1)]\\&=1-\left(e^{-3.5}+e^{-3.5}(3.5)\right)=0.8641\end{aligned}$$

Expectation value and variance

If X is a discrete r.v such that $X \sim \mathrm{P}_0(\lambda)$, then $E(X) = \lambda$ *and* $Var\ X = \lambda$

Example

If $X \sim \mathrm{P}_0(\lambda)$ with standard deviation 3. Find

a) E(X)
b) P(X<4)

Solution

a) Since standard deviation $\sigma = \sqrt{Var\ X}$, we have $\sqrt{Var\ X} = 3 \Rightarrow Var\ X = 9 = \lambda$

Consequently, since $X \sim P_0(\lambda)$, $E(X) = Var\ X = 9$

Thus the pdf of X is given by $P(X = x) = \dfrac{e^{-9}\,9^x}{x!}, x = 0, 1, 2, \ldots..$

b) $P(X < 4) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$

$$= e^{-9} + e^{-9} * 9 + \frac{e^{-9} * 9^2}{2!} + \frac{e^{-9} * 9^3}{3!} = 0.02122....$$

Relationship between Binomial and Poisson Distributions

- In the binomial distribution, if n is large while the **probability** p of occurrence of an event is **close to zero**, so that $q = 1-p$ is **close to one**, the event is called a **rare event**.
- In practice, we shall consider an event as **rare** if the number of trials is at least 50 ($n \geq 50$) while np is less than 5.
- For such cases, the binomial distribution is very closely approximated by the Poisson distribution with $\lambda = np$.
- This is to be expected on comparing the equations for the means and variances for both distributions.
- By substituting $\lambda = np$, $q \approx 1$ and $p \approx 0$ into the equations for the mean and variance of a binomial distribution, we get the results for the mean and variance for the Poisson distribution.

Exercise for Binomial distribution

1

The probability that a patient recovers from a rare blood disease is 0.4. If 15 people are known to have contracted this disease, what is the probability that

1. at least 10 survive
2. from 3 to 8 survive
3. exactly 5 survive?

Exercises for poison distribution

1

During a laboratory experiment, the average number of radioactive particles passing through a counter in 1 millisecond is 4. What is the probability that 6 particles enter the counter in a given millisecond?

2

Ten is the average number of oil tankers arriving each day at a certain port. The facilities at the port can handle at most 15 tankers per day. What is the probability that on a given day tankers have to be turned away?

Exercises cont...

3

Suppose we are interested in the number of people who visit the clinic in city "X" in a given year among the total population say 5000,000, and let the probability that some one in the city visits the clinic is 0.00001. The mean number of people from the example above would be $np = 5000000 * 0.00001 = 50$ which is also the variance. For this example calculate:

1. The probability that no one in this population visits the clinic in the a given year
2. The probability that less than 5 people visits the clinic in the a given year

Continuous Probability Distributions

1. Continuous Uniform Distribution

The density function of the continuous uniform random X on the interval $[a, b]$:

$$f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise.} \end{cases}$$

The cumulative density function $F(X)$ is

$$F(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x \leq b \\ 1 & x > b. \end{cases}$$

The moment generating function of the uniform distribution is given by

$$G_x(t) = E(e^{tX}) = \frac{e^{tb} - e^{ta}}{t(b-a)}.$$

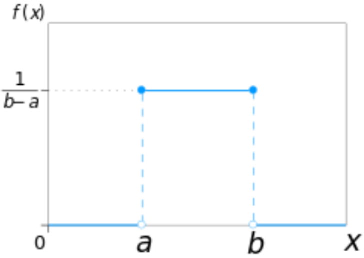


Fig4.2.Graphic of $f(x)$

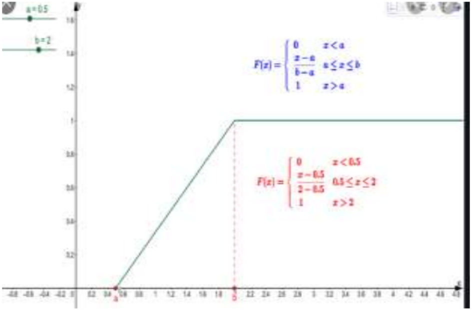


Fig4.3: Graphic of $F(x)$

Proposition

The mean and variances of the uniform distribution

$$E(x) = \frac{a+b}{2} \qquad \sigma^2 = \frac{(b-a)^2}{12}.$$

Example

Suppose large conference room for a certain company can be reserved than 4 hours. However, the use of the conference room is such that both long and short conferences occur quite of often. In fact, it can be assumed that length X of a conference has a uniform distribution on the interval $[0, 4]$.

1. What is the probability density function?
2. What is the probability that any given conference lasts at least 3hours.
3. What are the mean and variance.

Answer

1. The appropriate density function for the uniformly distributed random variable X in this situation is

$$f(x) = \begin{cases} \frac{1}{4} & 0 \le x \le 4 \\ 0 & \text{otherwise.} \end{cases}$$

2. The probability that any given conference lasts at least 3hours is given as

$$P(X \ge 3) = \int_3^4 \frac{1}{4}\, dx = \frac{1}{4}.$$

3. The mean and variance values are calculated as

$$E(x) = \frac{a+b}{2} = \frac{0+4}{2} = 2$$
$$\sigma^2 = \frac{(b-a)^2}{12} = \frac{(4-0)^2}{12} = \frac{16}{12} = \frac{4}{3}.$$

2. Exponential distribution.

The continuous random X has an exponential distribution with parameter λ,if its density function is given by

$$f(x,\lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0. \end{cases}$$

The cumulative density function $F(X)$ is given by:

$$F(x) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0 \\ 0 & x < 0. \end{cases}$$

The moment generating function of the exponential distribution is given by

$$G_x(t) = E(e^{tX}) = \frac{\lambda}{\lambda - t}.$$

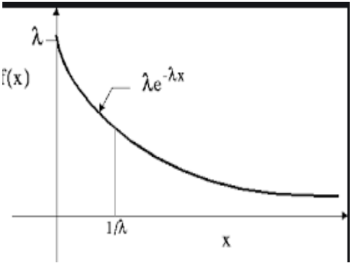


Fig4.4: pdf for exponential distribution

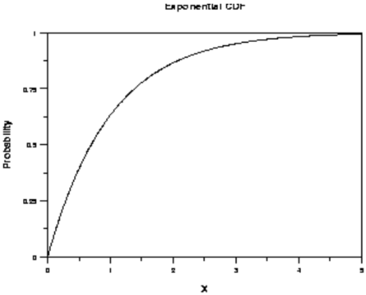


Fig4.5 :CF for exponential distribution

Proposition

The mean and variance of the exponential distribution are

$$E(X) = \frac{1}{\lambda}; \qquad \sigma^2 = \frac{1}{\lambda^2}.$$

Example

Suppose that a system contains a certain type of component. Whose time n year, to failure is given by X. The random variable X is modeled nicely by the exponential distribution with mean time to failure exponential distribution is $\lambda = \frac{1}{5}$. If 5 of these components are installed in different systems, what the probability that at least 2 are still functioning at the end of 8 years.

Answer:

The probability density function is given by:

$$f(x, \lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0. \end{cases} = \begin{cases} \frac{1}{5} e^{-\frac{1}{5}x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

The probability that a given component is still functioning after 8 years is given

$$P(X > 8) = \frac{1}{5} \int_{8}^{+\infty} e^{-\frac{1}{5}x}\, dx = e^{-\frac{8}{5}} = \approx 0.2.$$

3. The Normal Distribution

Def

It is a special distribution that we will use just about every day for a continuous random variable, has the following properties:

Properties

1. It can take on any value (not just integers, as do the binomial and Poisson distribution)
2. It is symmetric about the mean, μ
3. The standard deviation of the distribution is symbolized by σ, which is the horizontal distance between the mean and the point of inflection on the curve.
4. It approaches the horizontal axis on both the left and right side without touching, that is the x-axis is a asymptote.
5. It is bell shaped with transition points one standard deviation from the mean.

Formula

Given a normal random variable X, with mean μ and variance σ^2 that can take on any value between negative and positive infinity $(-\infty \quad \text{to} + \infty)$, normal distribution formula is as follows:

$$P(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}, -\infty < x < \infty$$

where $\pi = 3.14159...$ and $e = 2.71828...$

Figure

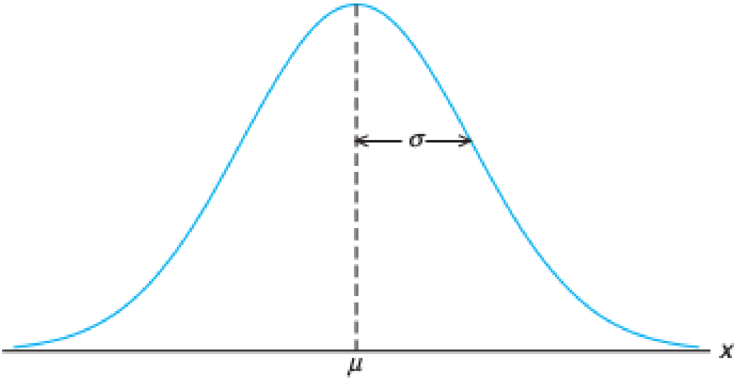


Figure: Normal curve

Properties cont ...

1. Approximately 68% of the data points lie within one standard deviation of the mean.
2. Approximately 95% of the data points lie within two standard deviations of the mean.
3. Approximately 99.7% of the data points lie within three standard deviations of the mean.
4. The total area under the curve and above the horizontal axis is equal to 1.
5. Since it is symmetrical distribution, half the area is on the left of the mean and half is on the right

The Standard Normal Distribution

Def

The Standard Normal distribution follows a normal distribution and has mean 0 and standard deviation 1.

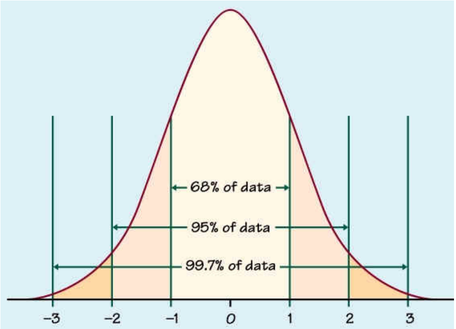


Figure: Standard normal distribution

Note

Notice the Figure 2 is perfectly symmetric about 0. If a distribution is normal but not standard, we can convert a value to the Standard normal distribution Z score by finding first as how many standard deviations away the number is from the mean. The number of standard deviations from the mean is called the *z-score* and can be found by the formula:

$$z = \frac{x - \mu}{\sigma}$$

The Z Score and Area

Often we want to find the probability that a z-score will be less than a given value, greater than a given value, or in between two values. To accomplish this, we use the Table from the textbook and a few properties about the normal distribution.

Finding the probability using the z-score

The z-score or p-value

- Each z-score associated with a probability, or p-value, that tells you the likelihood if values below that z-score occurring.
- If you convert an individual value into a z-score, you can then find the probability of all values up to that value occurring in a normal distribution.

Example

To find the probability of SAT scores in your sample exceeding 1380, you first find the z-score. The mean of our distribution is 1150, and the standard deviation is 150, the z-score tells you how many standard deviations away 1380 is from the mean.

$$z = \frac{x - \mu}{\sigma} = \frac{1380 - 1150}{150} = 1.53.$$

For a z-score of 1.53, the p-value is 0.937. This is the probability of SAT scores being 1380 or less (93.7%).

Example 1

Given a standard normal distribution, find the area under the curve that lies

1. to the right of $z = 1.84$ and
2. between $z = -1.97$ and $z = 0.86$.

Solution

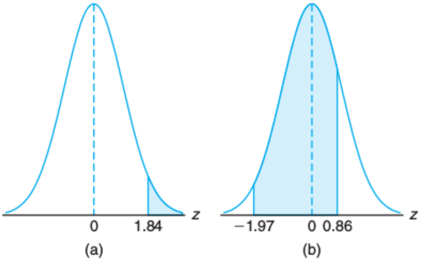


Figure: Area under the curves for example 1.

Solution cont ...

(a) The area in Figure 3(a) to the right of $z = 1.84$ is equal to 1 minus the area in Table A.3 to the left of $z = 1.84$, namely, 1 - $0.9671 = 0.0329$.

(b) The area in Figure 3(b) between $z = -1.97$ and $z = 0.86$ is equal to the area to the left of $z = 0.86$ minus the area to the left of $z = -1.97$. From Table A.3 we find the desired area to be 0.8051 - $0.0244 = 0.7807$.

Example 2

Given a standard normal distribution, find the value of k such that

1. $P(Z > k) = 0.3015$ and
2. $P(k < Z < -0.18) = 0.4197$.

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solution

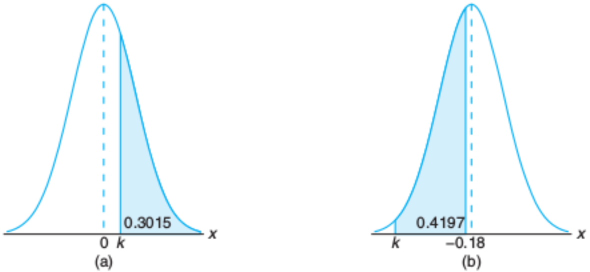

Figure: Area under the curves for example 2.

Example 2 cont...

Solution cont ...

(a) In Figure 4(a), we see that the k value leaving an area of 0.3015 to the right must then leave an area of 0.6985 to the left. From Table A.3 it follows that $k = 0.52$.

(b) From Table A.3 we note that the total area to the left of - 0.18 is equal to 0.4286. In Figure 4(b), we see that the area between k and -0.18 is 0.4197, so the area to the left of k must be 0.4286 - 0.4197 $= 0.0089$. Hence, from Table A.3, we have $k = -2.37$.

Example 3

Given a random variable X having a normal distribution with $\mu = 50$ and $\sigma = 10$, find the probability that X assumes a value between 45 and 62.

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Given a random variable X having a normal distribution with $\mu = 50$ and $\sigma = 10$, find the probability that X assumes a value between 45 and 62.

solution

The z values corresponding to $x_1 = 45$ and $x_2 = 62$ are

$$z_1 = \frac{45 - 50}{10} = -0.5 \quad \text{and} \quad z_2 = \frac{62 - 50}{10} = 1.2 \quad \text{respectively}$$

Therefore, $P(45 < X < 62) = P(-0.5 < Z < 1.2)$.

Figure

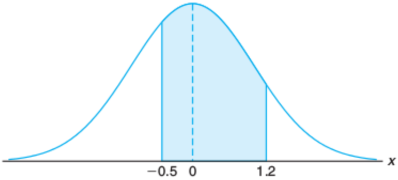

Figure: Area under the curve for example 3.

Cont...

$P(-0.5 < Z < 1.2)$ is shown by the area of the shaded region in Figure 5. This area may be found by subtracting the area to the left of the ordinate $z = -0.5$ from the entire area to the left of $z = 1.2$. Using Table A.3, we have $P(45 < X < 62) = P(-0.5 < Z < 1.2) = P(Z < 1.2) - P(Z < -0.5) = 0.8849 - 0.3085 = 0.5764$.

Exercise 1

The average salary for first-year teachers is $\$27,989$. Assume the distribution is approximately normal with standard deviation $\$3250$.

1. What is the probability that a randomly selected first-year teacher makes between $\$20,000$ and $\$30,000$ each year?
2. What is the probability that a randomly selected first-year teacher has a salary less than $\$20,000$?

Exercise 2

Test scores for a given class are normally distributed with a mean of 80 and standard deviation of 12, with several over 100 (with bonus points). Find the probability that a randomly selected student from this class has a test score that is...

1. less than 92
2. between 80 and 90
3. greater than 70

Exercise 3

Find the number b that satisfies:

1. $P(z < b) = 0.9664$
2. $P(z < b) = 0.3050$
3. $P(z > b) = 0.0078$

Relationship between Binomial and Normal Distributions

- If n is **large** and if **neither** p **nor** q **is too close to zero**, the binomial distribution can be closely approximated by a normal distribution with standardized random variable given by $Z = \frac{X - np}{\sqrt{npq}}$
- Here X is the random variable giving the number of successes in n Bernoulli trials and p is the probability of success.
- The approximation becomes better with increasing n and is exact in the limiting case.
- In practice, the approximation is very good if both np **and** nq **are greater than 5.**
- The fact that the binomial distribution approaches the normal distribution can be described by writing

$$lim_{n \to \infty} P\left[a \leq \frac{X - np}{\sqrt{npq}} \leq b\right] = \frac{1}{\sqrt{2\pi}} \int_a^b e^{-\frac{u^2}{2}}\, du.$$

- In words, we say that the standardized random variable $\frac{(X-np)}{\sqrt{npq}}$ is asymptotically normal.

Exercise

Find the probability of getting between 3 and 6 heads inclusive in 10 tosses of a fair coin by using (a) the binomial distribution and (b) the normal approximation to the binomial distribution.

(a) Let X have the random variable giving the number of heads that will turn up in 10 tosses. Then

$$P(X=3)=\binom{10}{3}\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^7=\frac{15}{28} \qquad P(X=4)=\binom{10}{4}\left(\frac{1}{2}\right)^4\left(\frac{1}{2}\right)^6=\frac{105}{512}$$

$$P(X=5)=\binom{10}{5}\left(\frac{1}{2}\right)^5\left(\frac{1}{2}\right)^5=\frac{63}{256} \qquad P(X=6)=\binom{10}{6}\left(\frac{1}{2}\right)^6\left(\frac{1}{2}\right)^4=\frac{105}{512}$$

Then the required probability is

$$P(3 \le X \le 6)=\frac{15}{28}+\frac{105}{512}+\frac{63}{256}+\frac{105}{512}=0.7734$$

Cont...

(b) Treating the data as continuous, it follows that 3 to 6 heads can be considered 2.5 to 6.5 heads. Also, the mean and the variance for the binomial distribution is given by $\mu = np = (10)\left(\frac{1}{2}\right) = 5$ and $\sigma = \sqrt{npq} = \sqrt{(10)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)} = 1.58.$

Cont...

$$2.5 \text{ in standard units} = \frac{2.5-5}{1.58} = -1.58$$

$$6.5 \text{ in standard units} = \frac{6.5-5}{1.58} = 0.95$$

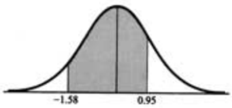

Figure 5-6

$$\begin{aligned}
\text{Required probability} &= (\text{area between } z=-1.58 \text{ and } z=0.95) \\
&= (\text{area between } z=-1.58 \text{ and } z=0) \\
&\quad + (\text{area between } z=0 \text{ and } z=0.95) \\
&= 0.4429 + 0.3289 \\
&= 0.7718
\end{aligned}$$

Relationship between Poisson and Normal Distributions

- Since there is a relationship between the binomial and normal distributions and between the binomial and Poisson distributions, we would expect that there should be a relation between the Poisson and normal distributions.
- This is in fact the case. We can show that if X is the following Poisson random variable

$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!} \quad \text{and} \quad Z = \frac{X - \lambda}{\sqrt{\lambda}}$$

as the corresponding standardized random variable. Then

$$lim_{n\to\infty} \quad P\left[a \leq \frac{X-\lambda}{\sqrt{\lambda}} \leq b\right] = \frac{1}{\sqrt{2\pi}} \int_a^b e^{-\frac{u^2}{2}}\, du.$$

- That is means that the Poisson distribution approaches the normal distribution as $n \to \infty$ or $\frac{(X-\lambda)}{\sqrt{\lambda}}$ is asymptotically normal.

Sampling methods

Probability sampling method (Random)

The best way to ensure that a sample will lead to reliable and valid inferences is to use probability samples, in which the probability of being included in the sample is known for each subject in the population. The four commonly used probability sampling methods are:

- Simple random sampling;
- Systematic sampling;
- Stratified sampling;
- Cluster sampling.

Simple random sampling

A **simple random sample** (lottery method) is one in which every subject has an equal **probability of being selected** for the study. The recommended way to select a simple random sample is to use a table of random numbers or a computer-generated list of random numbers.

Simple random sample is the simplest form of probability sampling. To select a simple random sample you need to make a numbered list of all the units in the population from which you want to draw a sample or use an already existing one (sampling frame). Then;

* Decide on the size of the sample;
* Select the required number of sampling units, using a lottery method or a table of random numbers.

Systematic random sampling

- It is an alternative to simple random sampling that is useful in some cases.
- The items or individuals of the population are arranged in some order. A random starting point is selected (by lottery) and then every k^{th} member of the population is selected.
- The k is determined by dividing the total number of items in the sampling frame by the desired sample size.

For example, if there are $N = 1000$ stores along Fifth avenue and we want to select $n = 100$ stores in the sample, $k = \frac{N}{n} = 10$. We shuffle only the first k, and select one, say 4. Now on we systematically select stores by adding $k, 2k, 3k, 4k$ etc to 4. So a systematic sample will have store number $4, 14, 24, 34, 44, 54$ etc.

Stratified random sampling

- A **stratified random sample** is one in which the population is first divided in to relevant **strata** (subgroups), and a random sample is then selected from each stratum proportionally.
- Characteristics used to stratify should be related to the measurement of interest, in which case stratified random sampling is the most efficient.
- A population is first divided into subgroups, called strata, and a sample is selected from each stratum. (e.g., 70% males, 30% females).
- If a sample of 10 is selected, (n=10) 70% of $n = 7$, so select 7 males and 3 females. In general, N=population size, N_1=stratum 1(female), N_2= stratum 2 (males), n=sample size desired.
- Sample should have $\dfrac{N_1}{N}n$ individuals from stratum 1 and so on.

Cluster random sampling

- It is a survey procedure in which the sampling units consist of a group of elements known as a **cluster**.
- Information on clusters is used in making inferences about a population.
- The scheme of selection of these clusters could be simple random sampling or systematic sampling.
- Clusters are usually formed by grouping units which can be conveniently observed together.
- These could either be mutually exclusive or overlapping.
- In practice, however, it is more convenient to use clusters that are non-overlapping.
- In cluster sampling, the first task is to specify the appropriate clusters.

Major Advantages of Cluster Sampling

The three major advantages of using cluster sampling in some surveys are:

1. The sampling frame from which the list of individual elements can be obtained may not be available or may not be easily obtained, but that of the clusters may be available and can be obtained easily.
2. It is more expensive to construct a list of elements in a population compared to a list of clusters. Greater costs may be incurred in taking a sample of elements as compared to clusters of elements.
3. The use of cluster sampling usually requires less personnel and takes a shorter time to obtain results

Non random sampling

Non-probability samples are those in which the probability that a subject selected is unknown.

Convenience sampling

It is exactly what the name says. That is one chooses a sample that is convenient. Examples;

1. When checking oranges packed in boxes, only oranges from the top layer are inspected,
2. Examining plants that are easily accessible, for example, found near a road.
3. Sampling patients coming to a specific clinic on a specific day. That is if the clinic is in the morning you may miss children that are at school.

Quota sampling

It is often used for opinion surveys. That is each interviewer is told to collect opinions from a specified number of items.

1. Select quota based on known characteristics of the population.
2. For example, age breakdown of a population Sample would be proportional to the size of each age segment.
3. You do not know whether the individuals that you select from each characteristic are representative of that characteristic.
4. Systematic bias can result.